

Quiet Neurons: a BDH quietens on what it just learned, not on what is merely predictable

DataForge 2026, Pathway track · concept **Sparse Non-Negative Activations** · Nilay Toshniwal, solo · artifact nilaymastaadmi.github.io/quiet-neurons · code github.com/nilaymastaadmi/quiet-neurons

THE DESIGN PRESSURE

A dense Transformer evaluates every feed-forward unit for every token, hard or not. Activation sparsity is the counter-proposal. But Transformer sparsity is already *emergent*, appearing during training rather than by design [3]; it needs a hard rectifier, because only that yields exact zeros [4]; and *which* units fire is input-dependent, not merely how many [5]. Sparsity by itself therefore distinguishes nothing. What distinguishes an architecture is what its sparsity is a function of.

WHAT BDH CHANGES

Dragon Hatchling (BDH) [1] makes it structural. In the BDH-GPU formulation Pathway open-sources, every layer computes two rectified (ReLU) vectors and multiplies them elementwise, and a product of two sparse non-negative vectors is sparser than either. *Active* throughout means a strictly non-zero entry in that product. Attention is the second change: query and key are the same tensor, the code asserting K is Q , so the score matrix is a causally-masked Gram matrix of neuron activations with **no softmax** in it. Attention is therefore a similarity between firing patterns, positions whose neurons fired alike bind to each other. That is the paper's Hebbian reading, in one line of tensor code.

THE CLAIM, AND THE CORRECTION IT NEEDED

Section 6.4 reports that "fewer neurons are active ... for more predictable input signals": at layer 2 of a 65,536-neuron model, 4.0–7.5% active while memorising a new word against about 2.5% while repeating it [1, Fig. 14]. We reproduced that protocol on three models trained here from the public code and it holds, surviving a 32-fold shrink. **In aggregate the paper is right: predictable text is quieter.** What follows sharpens that result rather than contradicting it.

Model (neurons / parameters)	Memorising	Repeating	Ratio	Evidence class
ours, 2,048 / 397k	8.24%	5.76%	1.43×	reproduced here
ours, 8,192 / 3.15M	7.18%	3.64%	1.97×	reproduced here
ours, 16,384 / 6.30M	8.20%	4.38%	1.87×	reproduced here
Pathway, 65,536 [1]	4.0–7.5%	~2.5%	1.6–3.0×	developer-reported

Each row is 1,280 sequences over five pinned samples, spread near 0.005. The same measurement then narrows what the result can mean. Two blocks of one sequence are both predicted almost perfectly and differ two and a half fold in activity: the 13-letter warm-up runs at **9.1%** of layer 2 active at 0.0005 nats of mean surprise (a nat is the natural-log unit of cross-entropy; zero means perfectly predicted), the 56 repeated letters at **3.6%** at 0.008 nats. At the extremes the gap is 3.9x. The warm-up is identical in every training sequence, so it lives in the **weights**; the repeated word is new each sequence, so it can only come from the **context**. What separates them is not predictability but **where the knowledge came from**: the activation level tracks parametric against in-context memory. We show that it tracks; we do not show why. A learner falsifies this in under a minute by injecting an unpredictable letter mid-repetition: activity rises at that letter, +10% to +20% by word, and is unchanged to nine decimals at the letter before it, because the model reads only leftwards.

EVIDENCE, LABELLED

REPRODUCED HERE: the three ratios, the counterexample, the surprise-injection check, and parity of the in-browser model against PyTorch (logit max difference 4.1e-5, eleven of twelve sparsity series bit-identical). **DEVELOPER-REPORTED, NOT REPRODUCED:** the Figure 14 band; 97.4% on Sudoku Extreme, which Pathway's own repository states comes from their internal implementation and does not reproduce from the open code; 1B-to-600B pretraining-scaling experiments; and BDH-CQ's 29.5% pass@2 on ARC-AGI-1 at \$0.0007 per task [2]. **NOT MODELLED:** BDH-CQ's internals, whose "dimensions, exact update rules, and implementation details remain proprietary" [2]. **BDH-CQ has no direct role in this concept;** inventing one would be worse than saying so.

LIMITATIONS, MATURITY, AND WHAT WOULD SETTLE THEM

The effect is not uniform with depth: layer 0 runs *backwards* at all three sizes (0.82, 0.79, 0.86) and layer 3 is flat below 16,384 neurons, weakly positive (1.11×) there. Scaling is **not monotonic** — 16,384 measures below 8,192 by twenty times the sampling spread — and we published the opposite before the third model finished. One confound stands: a single 4,000-step schedule cut by wall clock at 1,854, 1,917 and 2,309 steps. Close final losses (0.176, 0.175, 0.172) mean comparably trained, not controlled; three step-matched runs would settle it. These are toys, 0.4M to 6.3M parameters on a synthetic task, as the paper's own protocol is. Surprise and sparsity do *not* track per letter (8,192 neurons, layer 2: Pearson 0.29, Spearman 0.02), which killed a stronger claim. We count the elementwise product, the only quantity in the public code reaching the paper's band; the paper does not say which tensor it means, though that choice changes the effect's size, not its direction. Most importantly we establish the *signature*, not its cause: nothing here explains why context-held knowledge needs fewer neurons than weight-held knowledge. **MATURITY:** credible and outsider-checkable for this narrow property; the wider programme is earlier, its headline benchmarks developer-reported and its public comparisons stopping at GPT-2 scale.

[1] Kosowski, Uznański, Chorowski, Stamirowska, Bartoszkiewicz, *The Dragon Hatchling*, arXiv:2509.26507 (2025), §6.4 and Fig. 14. [2] Engdahl, Kosowski, Chorowski et al., *BDH-CQ*, arXiv:2608.09888 (2026). [3] Li, You, Bhojanapalli et al., *The Lazy Neuron Phenomenon*, arXiv:2210.06313 (2022), ICLR 2023. [4] Mirzadeh, Alizadeh, Mehta et al., *ReLU Strikes Back*, arXiv:2310.04564 (2023). [5] Liu, Wang, Dao et al., *Deja Vu*, arXiv:2310.17157 (2023), ICML 2023. — Reproduce every number: `python measure.py --ckpt checkpoints/bdh_n2048.pt --embd 64 --mult 32 --repeats 5`. Code and data MIT; bdh.py is Pathway's, unmodified, MIT.